

Kyuwon Weon

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EDUCATION

Northwestern University | Evanston, IL December 2026 (expected)
Master of Science in Robotics
Carnegie Mellon University (CMU) | Pittsburgh, PA May 2022
Bachelor of Science in Mechanical and Biomedical Engineering (University Honors)

SKILLS

Robotics: ROS 2, MoveIt, TF2, OpenCV, Gazebo, MuJoCo, Coppeliasim, RViz, SLAM, Pinocchio
Control: State-Space Modeling, Lagrangian Dynamics, PID, Forward/Inverse Kinematics, Motion Planning
Software: C++, Python, PyTorch, Linux (Ubuntu), Git, Unit Testing, MATLAB
Mechanical: SolidWorks, PTC Creo, FEA (ANSYS, COMSOL), Design for Six Sigma, DFM, Rapid Prototyping

EXPERIENCE

Alcon | *Medical Device Design Engineer* | Fort Worth, TX Jul 2022 - Jul 2025

- Led the full-lifecycle design and fabrication of a precision lens injector, utilizing rapid prototyping in Creo, 3D printing, and injection molding to iterate on mechanism geometry and validate DFM readiness
- Designed and assembled custom test fixtures to execute rigorous system-level testing, documenting performance data to drive design revisions and root-cause analysis

Alcon | *R&D Intern* | Belmont, CA Jun 2021 - Aug 2021

- Validated simulation fidelity by correlating FEA models in ANSYS with real-world sensor data, reducing the sim-to-real gap in deformation and optical response analysis

CMU Computational Bio-Modeling Lab | *Undergrad Researcher* | Pittsburgh, PA Sep 2020 - Dec 2020

- Generated synthetic training datasets using a C++ Finite Element Method (FEM) solver, converting raw simulation outputs into HDF5 format for ML integration
- Trained a 4-layer CNN in PyTorch to reproduce reaction-diffusion predictions, verifying the model's 300x acceleration over traditional methods

PROJECTS

Real-Time Safety-Critical Control Layer (CBF-QP) | Evanston, IL Jan 2026 - Present

- Architected a high-frequency safety filter for 7-DOF manipulation, utilizing Control Barrier Functions (CBF) and Quadratic Programming (QP) to mathematically guarantee forward invariance and collision avoidance
- Engineered Pinocchio-based kinematics pipeline with custom ROS 2 interface, currently scaling the framework to handle multi-primitive whole-body constraints and dynamic obstacle interaction

Real-Time APF Controller for 7-DOF Manipulator | Evanston, IL Dec 2025

- Implemented a reactive controller using Artificial Potential Fields (APF) to navigate dynamic environments without explicit path planning in MuJoCo and ROS 2
- Engineered global damping and gain scheduling algorithms to resolve high-frequency oscillations

PenPal – VLM-Guided Robot | *Motion Planner* | Evanston, IL Nov 2025 - Dec 2025

- Co-developed a VLM-driven autonomous writing system that answers questions using ROS 2, Python, and Git for version control and code reviews
- Implemented constrained motion planning in MoveIt to generate pen gripping and writing trajectories, while optimizing force/torque safety thresholds to permit contact-rich writing on a dynamic, human-held whiteboard
- Synchronized motion execution with the team's perception stack (RealSense/AprilTags) to compensate for real-time human-induced board movements

Autonomous Pen Grabbing Manipulator | Evanston, IL Sep 2025 - Sep 2025

- Programmed a PincherX-100 arm in Python to autonomously detect and grasp dynamic targets using OpenCV and RealSense